

Octo · arXiv 2405.12213 · RSS 2024

Empiricist

11+3

Smoke grids · tattoo numbers
800k/25 · +29% · 72% · 83/35/18 · 5%

Oct 14 paper club · VLA Architectures · Oct 16 BEHAVIOR nearby

Spine: open OXE GRP + diffusion chunks · underspecified horizons · NOT 2026 winner weights

Octo Model Team · Berkeley / Stanford / CMU / DeepMind · HF rail-berkeley/octo-*

Tattoo numbers board · Empiricist must memorize

800k / 25

OXE curated mix

vs RT-X ~350k

+29%

vs RT-1-X ZS

selected lang tasks

72%

FT avg Table I

vs scratch 20% / VC-1 15%

83 / 35 / 18

Diff / MSE / Disc

WidowX Small ablations

27M / 93M

Small / Base

not a 55B VLM

5%

Novel skill

Table VII · anti-hype

Empiricist priority matrix · GPU-honest

P0 Y	HF ckpt dry infer	Forward + sample_actions · VRAM fit 24 GB
P0 Y	Diffusion steps / chunk exec grid	Quality ≤ 20 steps? commit_seconds > raw H
P0 Partial	Language vs goal-image	Goals $\gg$ lang (+25 pp WidowX)
P1	FT vs scratch vs VC-1-style	One new domain ~100 demos
P1	New action head / wrist on-off	Block-wise adapters · wrist often hurts
P1	Bridge-only vs Octo mix init	Wide mix helps OOD FT
Defer	OXE TPU re-pretrain	$\gg 1k$ GPU-h · out of quiet scope
Defer	9-robot hardware suite	Multi-lab · not available here

Smoke grids · minimal Empiricist matrix

{ckpt: small|base} × {infer_steps: 5|10|20} × {exec_horizon: short|mid} × {cond: lang|goal}

ckpt	diff steps	H_exec	cond
octo-small 27M	5	short	lang
octo-base 93M	10	mid	goal
	20		

Then one FT-vs-scratch on a public stack

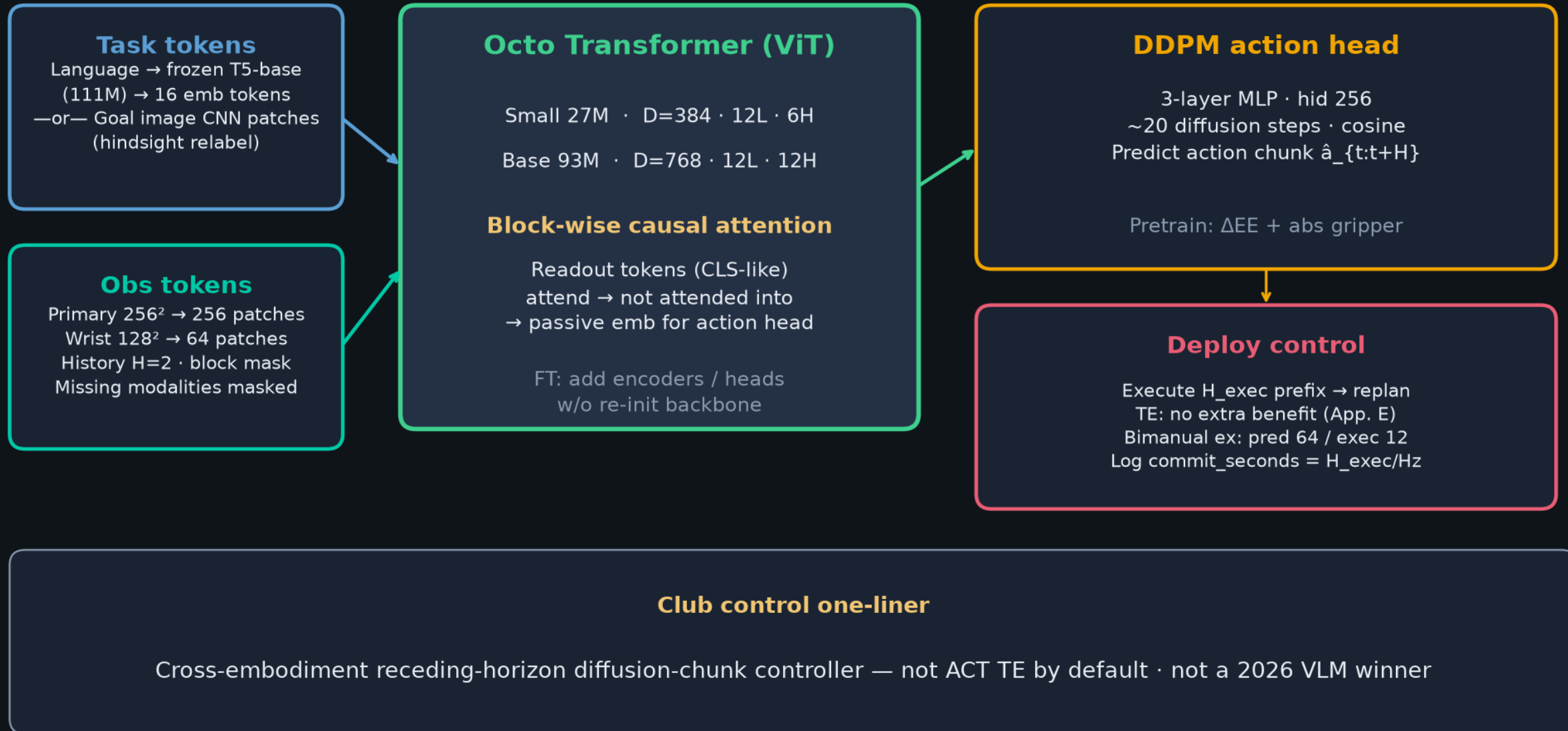
Log: GPU + RAM + ms + (ckpt, steps_diff, H_exec, Hz, cond, embodiment, seed)

Do NOT claim Table I 72% from Partial single-domain smoke

Pass bar: working open ckpt + qualitative Table II trends + one FT lift over scratch

Octo stack · ViT + frozen T5 + DDPM chunk head

Single backbone forward · light head denoises · receding-horizon execute (TE not helpful)



FT recipe card · operationalizable (rare honesty)

Data

~100 target demos

Steps

50k · cosine + warmup

Update

Full model (beat freeze)

Compute

~5 h · A5000 24 GB

Adapters

New cam / F/T / joint / 14-DoF

Result

Table I avg 72% vs 20%/15%

Domains: Insertion* (F/T) · Coffee · Baking · Pick-Up† · Coke · Bimanual† (*new obs †new action)

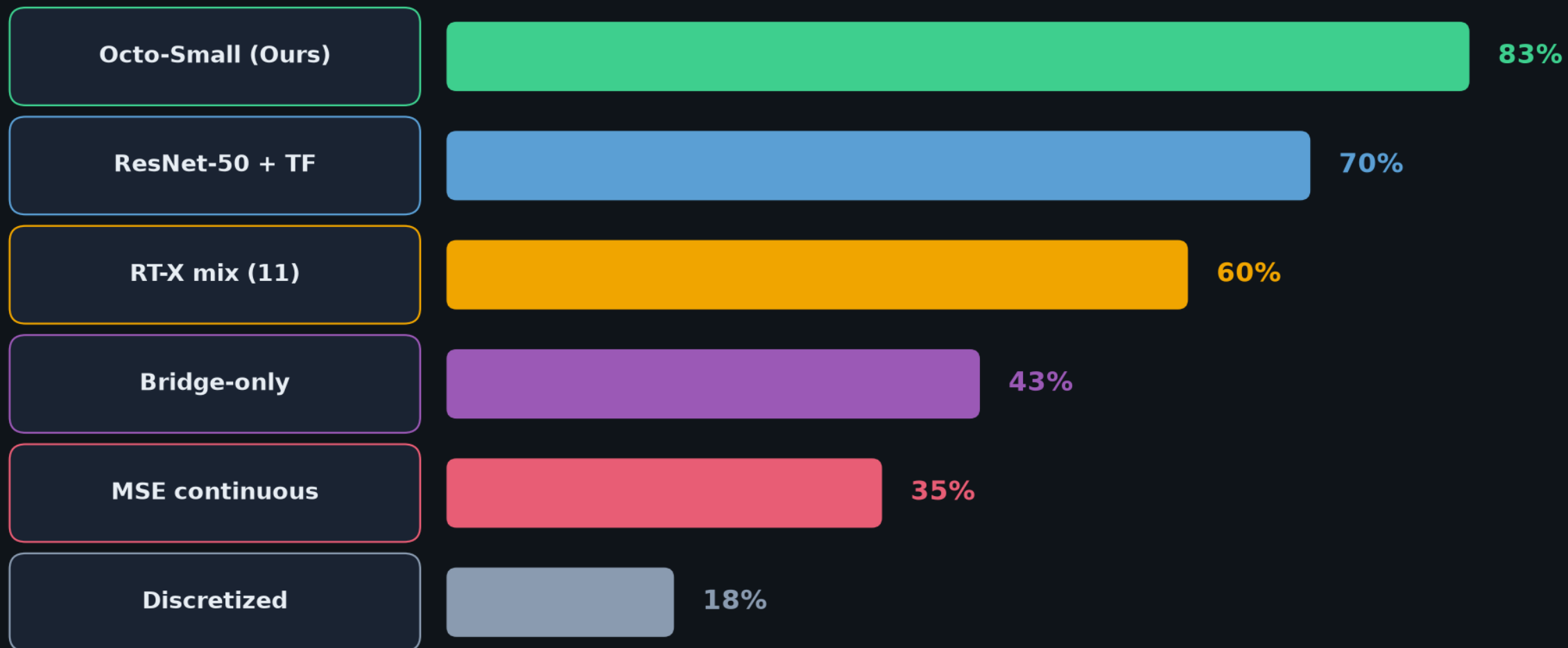
Underspecified horizons · motion FLAG

Main text: “several” future actions · only concrete: bimanual pred 64 / exec 12

System	Chunk object	Generative	Ensemble?	Typical commit
ACT	k×14 joints	CVAE	Yes TE	ensembled @ 50 Hz
Diffusion Policy	Tp actions	DDPM/DDIM	warm / Ta	Ta≈8 @ ~10 Hz
Octo (this)	action chunk	DDPM readout	No (paper)	receding prefix
π_0 / $\pi_{0.5}$	H actions	Flow matching	No	open-loop K @ 20-50
BEHAVIOR'25	H≈30-32	FM + corr.	soft inpaint	~30 Hz receding

Empiricist: always log (H_pred, H_exec, Hz, diff_steps) · commit_seconds = H_exec/Hz · TE not dead globally

Ablation board · Table II WidowX Octo-Small



Diffusion $\gg$ MSE/discrete · mix width 43 $\rightarrow$ 60 $\rightarrow$ 83 · ViT-first helps at pretrain scale

WidowX

achieved

language

e more

EdgeV2

zero-shot

as seen

skills.

objects,

e, and

precise

ains

remains

better

VC-1

etailed

C. Design Decisions for Generalist Robot Policy Training

We have demonstrated the effectiveness of Octo as a zero-shot multi-robot controller and as an initialization for policy finetuning. We next analyze the effects of different design decisions on the performance of the Octo policy. Concretely, we focus on the following aspects: (1) model architecture, (2) training data, (3) training objective, and (4) model scale. Unless noted otherwise, we perform all ablations on the Octo-Small model due to our compute budget.

Model architecture: Prior transformer-based policy designs typically encode input images with large ResNet-style [32] encoders and fuse the resulting image features with a comparatively small transformer [10, 67, 81, 17, 98, 61, 83]. Instead, we opt for a “transformer-first” architecture that uses very shallow CNN patch encoders and concentrates most of the parameters and FLOPS in the transformer backbone, similar to canonical vision transformer architectures [22]. In Table II

scratch as well as state-of-the-art pretrained visual representations. Each domain uses ~ 100 target demonstrations and the same finetuning hyperparameters. In each domain, success rates are averaged over 20 trials. *: New observation input (force-torque proprioception). †: New action space (joint position control).

placing, wiping a table with a cloth, and opening and closing drawers. For each robot, we selected two language tasks from the corresponding OXE dataset and performed 10 trials per task with varying initial conditions (details in Appendix F). The chosen tasks are “in-distribution” from the pre-training data, but the evaluation requires methods to generalize to new object positions, lighting conditions, backgrounds, and distractor objects. While all methods acted reasonably across tasks in the pretraining environments, we found that on average Octo had a 29% higher success rate than RT-1-X (35M parameters). For the WidowX and RT-1 Robot evaluations, we also compared to RT-2-X (55 billion parameters) [103] and found that Octo performed similarly.

Additionally, while RT-1-X and RT-2-X only support conditioning on language instructions, Octo also supports conditioning on goal images. We evaluated our model on the WidowX tasks using goal image conditioning and found that it achieved a 25% higher success rate than when evaluated with language conditioning. This is likely because goal images provide more information about how to achieve the task. In the BridgeV2 domain, we performed a fine-grained analysis of the zero-shot

		Aggregate Performance
DATA	Octo-Small (Ours)	83%
	RT-X dataset mix [67]	60%
	Single robot dataset (Bridge Data)	43%
POLICY	Discretized Action Prediction [67]	18%
	Continuous Action Prediction (MSE)	35%
ARCH	Resnet-50 + Transformer[67]	70%

TABLE II: **Model Ablations.** We achieve best performance when using the ViT architecture, diffusion action head, and wide training data mixture. All evaluations are performed on the WidowX setup. Success rates are averaged over 40 trials across two language-conditioned tasks and two goal-conditioned tasks.

C. Design Decisions for Generalist Robot Policy Training

We have demonstrated the effectiveness of Octo as a zero-shot multi-robot controller and as an initialization for policy finetuning. We next analyze the effects of different design

Generalization honesty · Table VII WidowX

In-distribution

85%

Novel objects

80%

Novel environment

40%

Novel skill

5%

flip / insert
ANTI-HYPE

Cite 5% novel skill whenever someone says “generalist zero-shot”

Objects transfer; environments degrade; skills near floor — prevents foundation-model overclaim

Fig. 2 · Model architecture — tokens → transformer → readout → action head

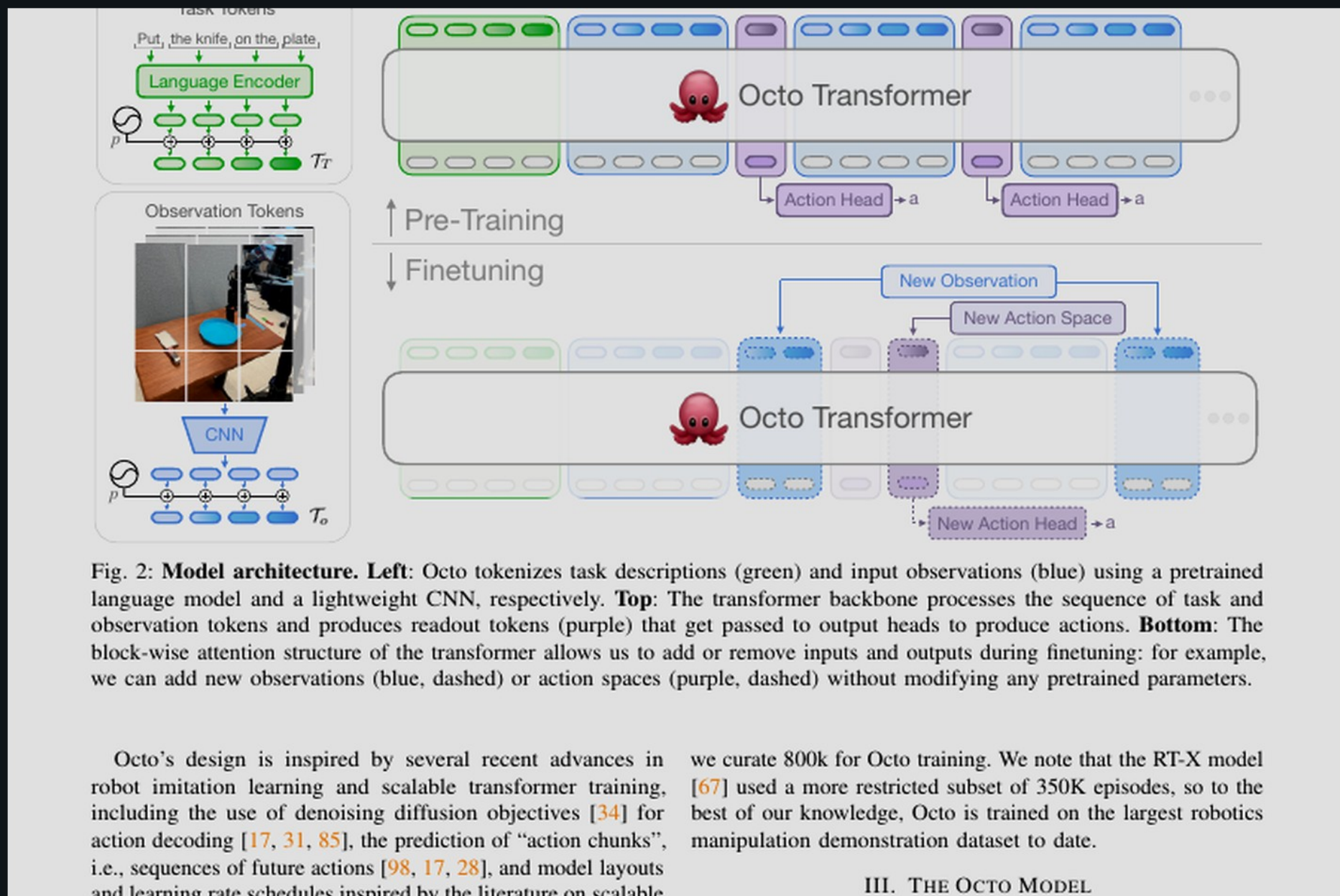


Fig. 2: **Model architecture.** **Left:** Octo tokenizes task descriptions (green) and input observations (blue) using a pretrained language model and a lightweight CNN, respectively. **Top:** The transformer backbone processes the sequence of task and observation tokens and produces readout tokens (purple) that get passed to output heads to produce actions. **Bottom:** The block-wise attention structure of the transformer allows us to add or remove inputs and outputs during finetuning: for example, we can add new observations (blue, dashed) or action spaces (purple, dashed) without modifying any pretrained parameters.

Octo's design is inspired by several recent advances in robot imitation learning and scalable transformer training, including the use of denoising diffusion objectives [34] for action decoding [17, 31, 85], the prediction of "action chunks", i.e., sequences of future actions [98, 17, 28], and model layouts and learning rate schedules inspired by the literature on scalable

we curate 800k for Octo training. We note that the RT-X model [67] used a more restricted subset of 350K episodes, so to the best of our knowledge, Octo is trained on the largest robotics manipulation demonstration dataset to date.

III. THE OCTO MODEL

Tattoo: 800k/25 · +29% · 72% · 83/35/18 · 27M/93M · novel skill 5%

P0: HF dry infer + diffusion/chunk grid

Budget FT ~24 GB · hours-1-2 days · Partial $\neq$ Table I

Q&A · keep the spine clear

Octo: An Open-Source Generalist Robot Policy

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<https://octo-models.github.io>

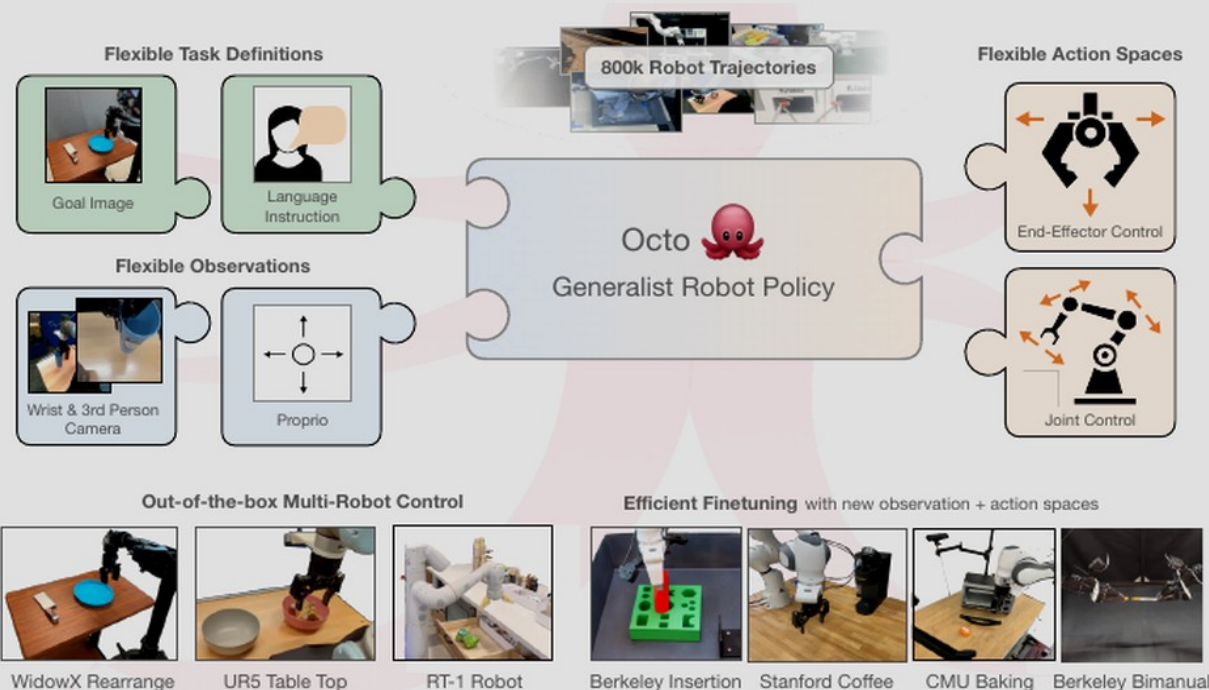


Fig. 1: We introduce Octo, an open-source, generalist policy for robotic manipulation. Octo is a transformer-based policy pretrained on 800k diverse robot episodes from the Open X-Embodiment dataset [67]. It supports flexible task and observation definitions and can be quickly finetuned to new observation and action spaces.

Abstract—Large policies pretrained on diverse robot datasets have the potential to transform robotic learning: instead of experiments across 9 robotic platforms, we demonstrate that Octo serves as a versatile policy initialization that can be effectively